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Still no smart service? A review of technical barriers to smart service adoption in the GenAI era

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ABSTRACT: OEMs are shifting from product-centric offerings toward smart services, but adoption is still hindered by technical development barriers. We conduct a systematic literature review of peer-reviewed studies with original industrial evidence and identify eight barrier categories across data, semantics, integration, governance and modularity. We map them onto a smart-service architecture and key analytics roles, and relate them to GenAI building blocks such as LLMs and knowledge graphs, outlining a research agenda for overcoming technical barriers towards scalable OEM smart service development.

KEYWORDS: product-service systems (PSS), artificial intelligence (AI), systematic literature review, technical barriers

1. Introduction

Original equipment manufacturers for special machinery (OEMs) today face intense competitive pressure and shifting customer expectations, driving a shift from traditional product-centric models toward integrated smart services (servitization and digitalization) (Reim et al., 2015). Smart services are bundles of connected products and data-driven services that leverage embedded sensors, software, and analytics to deliver predictive maintenance, real-time optimization, and remote monitoring (MF Filho et al., 2017). In this paper, we define smart services from the OEM perspective as technically engineered systems that couple cyber-physical assets with continuous data flows, enabling the systematic development, deployment, and evolution of new value-added services through a structured technical process (Beverungen et al., 2019).

Smart services offer OEMs a compelling economic proposition: they unlock new business models that generate recurring revenue streams with margins up to four times higher than one-off equipment sales, reduce revenue volatility, and foster stronger customer loyalty through differentiated offerings (Marquardt & others, 2017; Soellner et al., 2024; Vesco, 2024). Despite this promise, overall smart service adoption remains limited (Pezzotta et al., 2023; Smart Service Welt Working Group & acatech, 2015). To achieve a higher adoption and realize the economic potential of smart services, the robustness of the underlying technical development process is critical.

While existing literature has extensively examined business, organizational, and strategic barriers to smart service adoption, there is little systematic analysis of the technical barriers that OEMs face when engineering these systems. At the same time, the rapid emergence of generative artificial intelligence (GenAI) – including Large Language Models (LLMs) retrieval-augmented generation (RAG), code-generation agents and industrial knowledge graphs – presents new options for addressing such barriers, but their concrete impact on smart-service engineering remains unclear and requires a focused evaluation (Rane, 2023).

The contributions of this paper are threefold: First, barriers for technical development of smart services are identified and categorized through a systematic literature review. Second, the barriers are mapped

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onto a smart service architecture and relevant roles in the development process, to visualize their impact. Third, we link the identified barrier categories to concrete GenAI building blocks (e.g., LLMs, RAG, code-generation agents, industrial knowledge graphs) and outline a research agenda for integrating these technologies into smart service development.
Our results indicate that: (i) many barriers cluster around data management, semantics and integration, and (ii) these barriers disproportionately affect the role of the Data Engineer. We also identified that (iii) modular and contextual knowledge graphs enabled by GenAI technologies appear promising in resolving these barriers.

2. Methodology

For the methodology of this paper, we adopt the guidelines for preforming systematic literature reviews according to Kitchenham et al. (Kitchenham et al., 2007). The identification of the need for a review was achieved in Section 1. The research questions will be specified in Section 2.1, the review protocol will be developed in Section 2.2, the relevant research will be identified and selected in Section 2.3 and reported according to the PRISMA guidelines (Page et al., 2021) at the end of the same Section. Finally, the data will be extracted and synthesized with respect to the research questions in Section 3. In addition, Section 4 links the synthesized barrier categories to specific GenAI building blocks and discusses implications of integrating these technologies into the development process of smart services.

2.1. Research questions

While smart services have a tremendous potential for OEMs, adoption remains low (see Section 1). As a result, it is important to understand what technical barriers prevent smart service adoption. In addition, the current research of data-driven methods is dominated by GenAI technologies, therefore their effect on the barriers is also highly relevant. The research questions are therefore defined as follows:

1. What are barriers for smart service adoption of industrial companies, especially OEMs, concerning technical development?
2. How do the technical barriers to smart service adoption affect smart service subsystems and different roles in the development process?
3. Which GenAI building blocks are most promising to mitigate the identified technical barriers along the smart service architecture and across roles?

2.2. Research design

We choose Scopus as our primary database, since it offers extensive coverage of engineering, computer science, and management literature, indexes both peer-reviewed journals and leading conference proceedings and provides reliable metadata and export tools for reproducibility.
The search string was constructed by enforcing the inclusion of a keyword about the topic, the domain and the specific focus of this review. In addition, only literature from the last 10 years was considered to ensure that barriers still have relevance. We used the TITLE-ABS-KEY function in Scopus, that checks whether the contained keywords are present in the title, abstract or keywords. The search string was built as follows:

- **Topic:** *TITLE-ABS-KEY("smart service*" OR "product-service system*")*
- **Domain:** *TITLE-ABS-KEY("manufactur*" OR "assembly" OR "production" OR "engineering" OR "industr*" OR "machinery" OR "factory")*
- **Focus:** *TITLE-ABS-KEY("barrier*" OR "challenge*" OR "obstacle*" OR "limitation*" OR "hurdle*" OR "obstacle*" OR "survey" OR "review")*

We included only peer-reviewed journals and conference publications that focus on smart service development in the engineering and OEM context, specifically targeting the industrial production of complex machinery. To ensure relevance to our technical development perspective, we required that each study examine barriers of a technical nature, meaning issues that arise during the design, implementation, or integration of smart service capabilities, rather than barriers encountered by end-users or customers. We further restricted our sample to research articles that provide original empirical evidence from industry settings, such as surveys of engineering teams, case studies of pilot projects, field experiments, or action-research interventions. Studies that did not focus on the company's engineering and production processes or that lacked firsthand evidence on technical barriers were excluded.

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2.3. Identification and selection of research

1136 unique records were identified with the specified search string between the years of 2015 to 2025 (as of the 7th of July 2025). 1083 records were removed after screening the title and abstract. 303 of those were removed since they conducted research in a non-industrial domain or it was a domain-agnostic paper, 780 were removed since they did not contain original evidence from industry about barriers regarding the technical development of smart services. Most of the excluded papers either contained no original evidence from industry or presented a new method or approach, not focusing on the barriers themselves. The 53 articles that remained were checked for full-text availability, three full texts were not available to the authors. Finally, after the available full texts were assessed, a further 30 articles were excluded, since reading the full text revealed that they were too broad or did not contain information about technical barriers, contrary to what could have been assumed from reading the abstracts. Figure 1 offers an overview of the process of selecting the relevant literature.

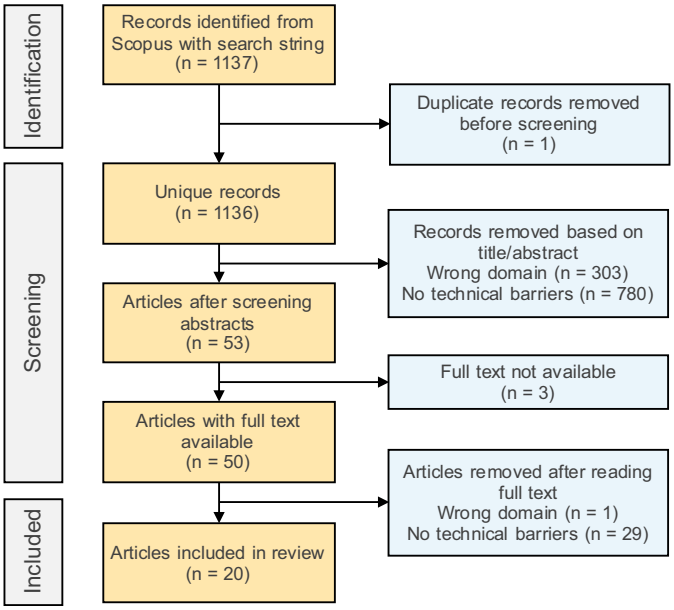


Figure 1. PRISMA-style flow chart of article identification and screening, based on (Page et al., 2021)

2.4. Data extraction, coding and GenAI mapping

For each included study, we extracted all statements that explicitly describe obstacles, limitations, or challenges encountered during the technical development of smart services. Based on this, the synthesis of barriers followed a structured qualitative coding process:

1. Barrier statements were extracted verbatim from each included study.
2. Similar formulations were normalized to common labels.
3. Labels were iteratively grouped into higher-level categories.
4. Category definitions were revised until internal coherence and cross-paper applicability were achieved.

To address RQ3, we used a targeted scoping linkage rather than a second full systematic review. We linked barrier categories to representative GenAI building blocks through barrier-specific searches and by selecting peer-reviewed approaches that (i) describe an implementable technical mechanism relevant to industrial data/software engineering and (ii) plausibly mitigate the respective barrier category. It should be noted that the resulting mapping is illustrative rather than exhaustive, which is sufficient for the purposes of this paper because it supports our goal of deriving plausible mitigation directions and a prioritized evaluation agenda, not of cataloguing all GenAI methods.

3. Findings

This section will present the data extraction and synthesis into a structured format. First, the extracted barriers will be coded and divided into categories to answer the first research questions in Section 3.1.

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Second, the barrier categories will be analyzed in terms of their effect on smart service systems and roles to answer the second research question in Section 3.2.

3.1. Extracted barriers

The literature was processed as described in Section 2.4, with the final categories presented in Table 1. In the following, all barrier categories will be explained in detail.

Table 1. Identified barrier categories, number of mentions in relevant literature and examples

Barrier Category	#Mentions	Examples
A. Engineering Methods, Tooling & Capability	10	(Åkesson et al., 2023; Arioli et al., 2025; Paliyenko et al., 2022)
B. Data Acquisition & Quality Governance	10	(Machchhar & Bertoni, 2021; Münch et al., 2022; Tawil et al., 2024)
C. Analytics & Algorithmic Limitations	8	(Münch et al., 2022; Tawil et al., 2024; Uuskoski et al., 2023)
D. Integration, Interoperability & Standards	8	(Heuchert et al., 2020; Pascher & Wulf, 2024; Wang et al., 2025)
E. Security, Privacy & Governance	7	(Heuchert et al., 2020; Mueller et al., 2023; Pascher & Wulf, 2024)
F. Context & Semantics	6	(Kamp et al., 2017; Paliyenko et al., 2023; Sala et al., 2022)
G. Infrastructure, Connectivity & Scalability	6	(Münch et al., 2022; Stoll et al., 2023; Wang et al., 2025)
H. Modularity, Architecture & Reconfigurability	5	(Larsen et al., 2019; Matschewsky et al., 2025; Pascher & Wulf, 2024)

A. Engineering Methods, Tooling & Capability: This category refers to the availability and maturity of methods, SDKs, simulation or digital twin support, and developer skills required to engineer smart services (Arioli et al., 2025). Gaps in design tooling, process guidance, and hands-on competence translate into slow, error-prone implementation (Åkesson et al., 2023). The absence of fit-for-purpose tools and practices inhibits systematic, repeatable development (Paliyenko et al., 2022).

B. Data Acquisition & Quality Governance: This category concerns what to measure, how to sample, and how to ensure completeness, reliability, and usability of data from heterogeneous sources (Münch et al., 2022; Tawil et al., 2024). Typical issues include sensor selection, sampling rates, missing or noisy data or unstructured records (Machchhar & Bertoni, 2021). Poor quality governance degrades downstream analytics and increases rework in pipelines.

C. Analytics & Algorithmic Limitations: This category includes shortcomings in modeling, explainability, robustness, and performance of algorithms deployed on industrial data (Münch et al., 2022; Tawil et al., 2024). Challenges arise from multi-modal signals, domain shift, and difficulties in translating model outputs into actionable insights. Insufficient accuracy or transparency undermines trust and delays deployment (Uuskoski et al., 2023).

D. Integration, Interoperability & Standards: This category covers incompatibilities in data models, interfaces, and protocols as well as the difficulty of integrating legacy systems and vendor-specific platforms (Pascher & Wulf, 2024). The lack of common standards and robust APIs impedes cross-system data exchange and reuse (Heuchert et al., 2020). Consequences include duplicated integration logic, fragile adapters, and limited scalability of solutions (Wang et al., 2025).

E. Security, Privacy & Governance: This category encompasses data protection, access control, regulatory compliance, and questions of data ownership and sharing (Mueller et al., 2023; Pascher & Wulf, 2024). OEMs must reconcile technical integration with legal and contractual constraints. Weak controls or unclear ownership structures block data flows and restrict the design space of services (Heuchert et al., 2020).

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F. Context & Semantics: This category captures the absence of machine-interpretable context, such as missing ontologies, unified namespaces, provenance, and domain semantics (Paliyenko et al., 2023; Sala et al., 2022). Without explicit semantics, data remain difficult to interpret, integrate, and trace across lifecycle stages (Kamp et al., 2017). The result is brittle analytics and duplicated modeling effort, especially when transferring insights across products and variants.

G. Infrastructure, Connectivity & Scalability: This category addresses runtime constraints such as network latency, bandwidth, real-time processing, storage/compute scaling, and cost-efficient operations (Münch et al., 2022; Wang et al., 2025). Bottlenecks emerge from distributed assets, edge–cloud coordination, and uneven connectivity (Stoll et al., 2023). These constraints limit the feasibility of time-critical services and increase total cost of ownership.

H. Modularity, Architecture & Reconfigurability: This category covers the architectural ability to compose, reuse, and evolve service modules across product variants and lifecycles (Matschewsky et al., 2025; Pascher & Wulf, 2024). Lack of modularization and variant management complicates maintenance and inflates engineering effort (Larsen et al., 2019). Reconfigurability deficits hinder rapid adaptation to changing requirements and operating contexts.

There was no discernible trend in how the number of mentions of each barrier evolved over the last 10 years, which may also be due to the low sample size of 20 research articles and 60 total barrier mentions.

An illustrative example for the stated barriers would be smart services based on remote troubleshooting of faults (Klein et al., 2025; Sala et al., 2024). An OEM might want to offer remote troubleshooting for special machinery using sensor data along with PLC/event logs, which needs to be analyzed in the context of maintenance tickets, manuals, configurations and similar documentation. In practice, engineers might face incomplete or noisy signals (barrier B), inconsistent naming and missing/non-traceable semantic context (barrier F), and substantial integration effort across systems and vendor interfaces (barrier D). Limited connectivity at customer sites and edge-cloud constraints further restrict the analysis (barrier B and G), which might be related to concerns regarding data security (barrier E). If the special machinery is able to be configured with different parts and functionalities, the services would also need to be designed in a modular fashion (barrier H) (Klein et al., 2025; Sala et al., 2024). These issues slow or even prevent development of the services.

3.2. Mapping of barriers to architecture and stakeholders

The second research question addresses the question of which systems, and which roles are affected by the identified barriers. Answering this question helps to pinpoint heavily affected systems and roles, narrows down the solution space and helps to map out migrations. For the purpose of this paper, we use the smart service reference architecture of Wegel et al. (2021), since it includes technical elements as well as a high level of detail for systems involved in the technical development process. Figure 2 offers an overview of a smart service reference architecture with annotations based on which elements are affected by the identified barriers.

Wegel et al.'s reference architecture extends the acatech layer model into a data-flow-oriented stack from Linked Physical Platforms (sensors, actuators, on-device processing) via Networking and IT-Infrastructure/Software Platforms up to the Service Platform and Business Model, with Security and Safety and Human–Machine Interface as cross-cutting concerns (Wegel et al., 2021). Our mapping shows that barriers are concentrated in the middle and top of the stack: data acquisition and quality issues (barrier B) appear mainly at the physical layer, but most barriers relate to data and resource management, analytics, and integration (barriers C, D, F, G) in the Networking, IT-Infrastructure and Software Platform layers, where heterogeneous data must be stored, enriched with context and semantics, and transformed into explainable insights (Kamp et al., 2017; Tawil et al., 2024; Wang et al., 2025). Near the top of the stack, barriers around governance, modularity and architecture design (barriers A, E, H) constrain how service processes, roles and adjacent IT systems (e.g. ERP) can exploit these data services (Åkesson et al., 2023; Matschewsky et al., 2025; Pascher & Wulf, 2024). This yields two key insights: (1) resolving mid-stack barriers around data management, semantics and integration is likely to have the largest leverage on smart-service development, and (2) governance- and architecture-related barriers propagate vertically across all layers, so they cannot be solved locally within individual components of the stack.

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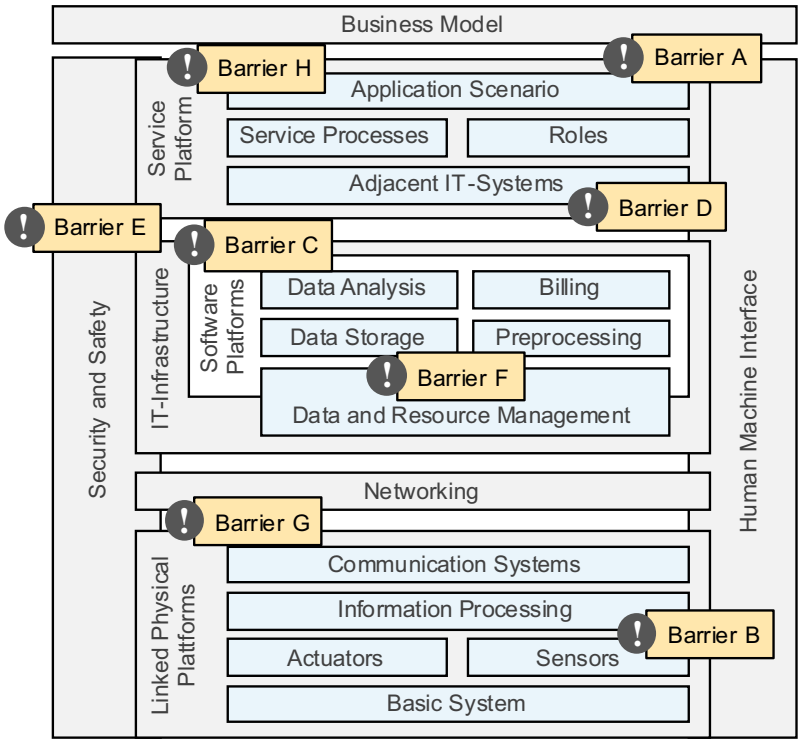


Figure 2. Identified barriers in relation to the smart service reference architecture based on Wegel et al. (Wegel et al., 2021) extended with identified barriers

Shifting focus from systems to development teams, Figure 3 shows which roles from a typical analytics department (role definitions adapted from (Kühn et al., 2018)) are affected by which barriers. If barriers are drawn between roles, it indicates that both roles are (partially) affected.

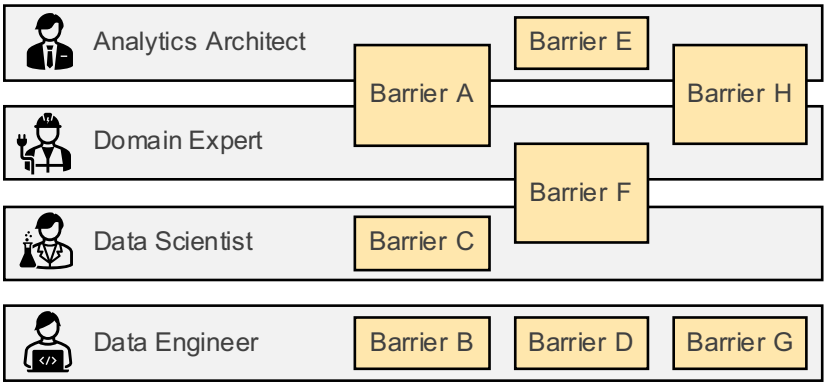


Figure 3. Mapping of roles of an analytics department to barrier categories

All four core roles in analytics-driven smart service development are affected by the identified barrier categories. Data Engineers are mainly associated with *Data Acquisition & Quality Governance (B)*, *Integration, Interoperability & Standards (D)*, and *Infrastructure, Connectivity & Scalability (G)*, reflecting their responsibility for deciding what to measure, connecting heterogeneous systems, and operating scalable data pipelines (Kamp et al., 2017; Münch et al., 2022). Data Scientists are primarily exposed to *Analytics & Algorithmic Limitations (C)* and *Context & Semantics (F)*, since they must build models on top of imperfect data and often lack a consistent semantic representation of machine states and events (Kamp et al., 2017; Wang et al., 2025). Domain Experts face *Engineering Methods, Tooling & Capability (A), Context & Semantics (F)*, and *Modularity & Reconfigurability (H)*, as they contribute operational knowledge, define service variants, and validate whether technical solutions match real-world needs (Larsen et al., 2019; Matschewsky et al., 2025). The Analytics Architect spans *Engineering Methods & Tooling (A)*, *Security, Privacy & Governance (E)*, and *Modularity & Reconfigurability (H)*, setting standards and guardrails across projects.

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Two insights emerge from this mapping. (1) all roles are systematically affected; technical barriers are not confined to "data people" but shape work across engineering, data science, and domain expertise. (2) data-engineering-related work is most affected: in our mapping, the Data Engineer role is linked to three barrier categories at full strength, roughly twice as many assignments as any other role. This suggests that many technical barriers manifest first as data-engineering challenges, making this role a critical leverage point for mitigation.

4. Discussion

Section 3 showed that technical barriers to smart services are pervasive across the architecture and roles, with a particular concentration in mid to top area of the stack with data management, semantics and integration, and a heavy burden on data-engineering work. RQ3 therefore asks which GenAI building blocks are most promising to mitigate these barriers. Table 2 summarizes representative approaches per barrier category.

Table 2. Exemplary GenAI-based approaches for mitigating technical barriers for selected categories

Barrier Category	GenAI-based Mitigation Approaches
A.	Natural language interfaces for knowledge (Hubauer et al., 2018), implementation assistance (Koziolek et al., 2024)
B.	Automated preprocessing agents (Naeem et al., 2024)
C.	Synthetic data (Boikov et al., 2021), feature enrichment using embeddings (Garofalo et al., 2018), multi-modal foundation models (Liu et al., 2025)
D.	LLM-based entity matching (Steiner et al., 2025), tool calling (Li et al., 2023)
F.	GenAI-supported knowledge graphs (Guo et al., 2024), unified embeddings (Garofalo et al., 2018)
H.	Automated architecture suggestions (Heissen et al., 2024), adaptive code generation (Koziolek et al., 2024)

A first cluster of approaches targets data and semantics–related barriers (B, D, F). Ontology-based databases and knowledge graphs have been proposed to structure heterogeneous lifecycle data (Hubauer et al., 2018). Typical sources include time-series sensor data, PLC/event logs, maintenance tickets, FMEAs and BOM/CAD exports, which are mapped into a common ontology describing machines, components, failure modes and operation conditions (Hubauer et al., 2018). Combining these heterogeneous data sources often involves manual modelling and population and limited their adoption (Guo et al., 2024). GenAI can partially alleviate this by using LLM-based entity matching and document understanding to populate and maintain knowledge graphs, and by linking heterogeneous sources via unified embeddings (Garofalo et al., 2018; Guo et al., 2024; Liao et al., 2025; Steiner et al., 2025). In practice, this can be implemented as a combination of a structured knowledge graph and an embedding index: LLM-based extractors add entities and relations to the graph, while embeddings support similarity search over documents and machine states. Natural-language interfaces on top of these graphs reduce the technical know-how needed to access context information (Hubauer et al., 2018; Li et al., 2023), which directly addresses the semantic bottlenecks observed in Section 3.

A second cluster addresses engineering methods, tooling and architecture (A, H, partly E). LLM-based assistants can support requirements refinement and implementation tasks (Hubauer et al., 2018; Koziolek et al., 2024), while architecture-suggestion systems and adaptive code-generation agents help to maintain modular architecture models and generate standardized integration and optimization code (Heissen et al., 2024; Koziolek et al., 2024; Li et al., 2023). Combined with tool-calling agents that interact with version control, CI/CD pipelines and external APIs, these techniques have the potential to reduce the coordination overhead identified for Analytics Architects and Domain Experts.

A third cluster targets analytics and runtime constraints (C, G). Synthetic data generation, feature enrichment with contextual embeddings and pre-trained multi-modal foundation models promise improved accuracy and robustness under diverse operating conditions (Boikov et al., 2021; Garofalo et al., 2018; Liu et al., 2025). Concretely, generative models such as variational autoencoders or diffusion models can be used to augment rare fault patterns, while time-series transformers and multi-modal

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encoders combine sensor trajectories with symbolic context from the knowledge graph (Fan et al., 2020; Klein et al., 2025; Liu et al., 2025). Agent-based preprocessing pipelines can continuously clean and transform incoming data streams, mitigating some of the scalability and data-quality problems faced by Data Engineers (Naeem et al., 2024).

However, employing GenAI also introduces new challenges. LLMs can hallucinate, leading to incorrect code, incorrect entities in knowledge graphs or misleading analytical interpretations, which is problematic in safety-critical industrial contexts. End-users, including machine operators at customer sites, need guidance on how to interpret and validate GenAI outputs (Huang et al., 2025). Moreover, many high-performing models are closed-source or resource-intensive, raising concerns about data privacy, intellectual property and dependency on external providers.

Overall, the evidence suggests that GenAI can help to reduce several types of barriers, but its benefits depend on robust semantics, governance and verification. This leads to three research directions: (1) design smart-service architectures that treat knowledge-graph/embedding layers and GenAI agents as core components; (2) develop assurance methods so GenAI outputs are traceable and auditable; and (3) empirically test whether GenAI reduces data-engineering effort and improves explainable, context-aware analytics. However, GenAI cannot remove structural barriers such as security/privacy/governance and infrastructure constraints, which require investments in robust systems and compliance. Modularity and process design also still rely on human expertise. Future work should therefore combine GenAI building blocks with non-GenAI improvements in infrastructure, processes, and governance.

5. Conclusion and outlook

This systematic review has provided an overview of technical barriers that stall smart service adoption in OEMs. Eight barrier categories were identified and synthesized across 20 primary studies, then mapped onto a smart-service architecture and core analytics roles. This mapping highlights that barriers concentrate in the mid to top-stack layers of data management, semantics and integration, and that data-engineering-centric work carries a disproportionate share of the burden, while governance and modularity issues propagate across all layers and stakeholders.

Building on these findings, we discussed how emerging GenAI approaches can aid in data preparation, semantic enrichment, integration and analytics design. Modular knowledge graphs, combined with GenAI building blocks (e.g., LLM agents, RAG, and embeddings), can help structure heterogeneous lifecycle knowledge and enable more capable and explainable analytics.

Future work could pursue two complementary directions. First, the review could be extended by broadening the literature base (e.g., additional databases and grey literature), by validating/refining the barrier categories through empirical studies, as well as interviews with OEMs to strengthen the evidence for the research agenda. Additionally, a systematic literature review of GenAI-based improvements could further structure solution patterns. Second, researchers and practitioners should design and evaluate modular frameworks that integrate knowledge-graphs, embedding layers and GenAI components with non-GenAI improvements in infrastructure, processes, and governance to test their improvement in task-specific metrics as well as the reduction of development time.

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